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## Can Correction Messages Reduce the Spread of Fake News on Social Media? The Impact of Information Updates on the Effectiveness of Corrections

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# Can Correction Messages Reduce the Spread of Fake News on Social Media? The Impact of Information Updates on the Effectiveness of Corrections

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## ABSTRACT

Although government agencies and fact-checking organizations issue correction messages to mitigate misinformation dissemination on social media, their efforts have been largely ineffective. Drawing on Information Manipulation Theory (IMT) and information extracted from over 84 million tweets, we examine the impact of correction messages and explore how different information updates affect their effectiveness. Our empirical approach differentiates between subpopulations of correction messages. The results support the existence of two distinct types: correction messages that significantly mitigate the spread of fake news—a 1 percent increase in “effective” messages decreases the spread of fake news by 0.89 percent; and those that amplify the misinformation—a 1 percent increase in “ineffective” messages increases the spread of fake news by 1.37 percent. Despite the higher count of “effective” correction messages, they are less potent than their “ineffective” counterparts. Our study extends the tenets of IMT to correction messages to identify effective platform interventions and highlight the efficacy of information updates in reducing the spread of fake news. Furthermore, it offers significant insights to social media platforms and fact-checking organizations on designing and deploying effective corrections.

## KEYWORDS

Social media; fake news; news corrections; message updates; information manipulation; online platforms

## Introduction

Fake news is of significant interest due to its impact on society. Combating fake news is important, as 50 percent of Americans view it as a much greater threat than either climate change or racism [1]. Unsurprisingly, social media users are routinely and consistently inundated with false information, which they then share, re-energizing its propagation [36]. As such, it has gained particular attention in the information systems (IS) community [32, 44], with several studies focusing on its mitigation.

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Potentially damaging rumors are sought to be tackled by issuing corrections, also known as corrective or correction messages. Such a message is often posted by someone who has fact-checked a rumor and holds an opposing view. Anyone exposed to a correction message can modify its content before reposting it. The modified message is called an updated correction or corrective message. Corrections are typically after-myth refutations in which information is often presented to counter the false claims [77]. However, these strategies can be ineffective or counterproductive if they backfire [52]. An effective correction reduces the propagation of misinformation, whereas an ineffective correction fails to reduce it by either amplifying the misinformation or having no effect in certain cases [45]. Although government agencies and fact-checking organizations respond promptly in refuting fake news, their campaigns have generally been ineffective and caused blowbacks [45]. Fragmented and inconsistent results of IS research further complicate the lacuna in correcting fake news. Furthermore, during crises, when information is lacking or ambiguous [2], public perceptions of the unknown “reality” are elastic [5]. However, as events unfold and new information becomes available, the public adjusts their beliefs and updates their information. Hence, understanding these updates and how they affect the diffusion of fake news is important in combating misinformation.

Existing research on the effectiveness of correction messages has often employed surveys, treated corrections as homogeneous, and overlooked the dynamic nature of communication. While individual-level interventions show promise in combating misinformation, their overall effectiveness has been primarily explored in controlled laboratory settings, with minimal exploration of their impact on social networks. This gap is likely due to several factors, including the feasibility of conducting field experiments, privacy, ethical considerations, and the complexities of tracing real-world information diffusion dynamics [51]. Furthermore, correction messages are of different types, and treating corrections as homogeneous can result in omitted variable bias. Distinguishing between unobserved types can help scholars, researchers, and practitioners identify and better understand effective corrections and their constituents. Additionally, although communication and beliefs are not static, previous research often treated them as such; this study addresses their (information updates) role in the effectiveness of corrections, an unexplored topic.

Our estimation strategy uses mixture models (see, e.g., [8, 50]) to identify subpopulations or types of correction messages within an overall population without labeled data. Each correction type can have a differentiated effect on the diffusion of fake news. Given the dynamic nature of information diffusion, we collected a unique panel dataset to identify how information updates influence responses to government and independent fact-checking campaigns. Understanding the inherent characteristics of effective and ineffective correction messages is an important step toward efficiently correcting misinformation. Our approach has several advantages. First, identifying different correction types comes from the diffusion of fake news and our data’s rich variation in correction messages. Second, the estimation incorporates the diversity of corrections by distinguishing between types and by using diffusion characteristics (i.e., updates) to model the probability of observing a particular type. Third, our approach allowed us to determine the effectiveness of different correction types on fake news propagation and how to tailor corrections accordingly. Finally, we argue that while correction updates affect correction effectiveness, the size and direction of the effects are moderated by media files.

The remainder of this paper is structured as follows. We first review the literature on fake news and corrections, types of corrections, and Information Manipulation Theory (IMT), and develop our research hypotheses. Next, we describe our data collection procedure and the empirical approach that models different correction types. We then present and analyze the estimation results, examining correction types and extending our analysis to investigate the influence of information update characteristics. The last section concludes the paper.

## Related Literature

### *Mitigating Fake News*

The impact of fake news and its spread are among the most extensively researched topics in IS [31], often focusing on human or machine interventions. Fake news is often created to evoke emotions [3, 12], and research suggests highly emotional individuals are more likely to react to, share, or suppress fake news [36, 43, 75]. Human and machine interventions often try to counter these emotional barriers by stimulating cognitive reactions to lessen their susceptibility to fake news. This is done, for example, by encouraging recipients to extend cognitive effort or binary thinking [61, 84] by prompting them to consider the information's credibility, questioning their belief in it [69], or their familiarity with related events [62].

Despite numerous intervention studies in IS research, the effectiveness of fake news corrections remains unclear. Most studies rely on individual-level interventions, which show promising results but are often difficult to replicate. Two main approaches to combating fake news exist: proactive (ex-ante) and reactive (ex-post). Machine- and human-based methods, such as detection [42] and information literacy [19, 55], are proactive approaches. Studies examining the effectiveness of these measures for combating fake news have often been inconsistent or inconclusive, mainly because machine-based interventions tend to rely on the design science paradigm to classify and differentiate fake news content from facts using machine learning models [85]. Researchers' attempts to surpass previous fake-content classification methods have yielded impressive results on paper but do not perform as well when deployed in real-world settings, possibly because of the nature of the data or white noise [31, 42]. The dynamic nature of fake news renders algorithmic interventions unreliable and impractical. At the same time, proactive human interventions, such as information literacy [19, 55], rely on small-scale experimental research, which may not accurately reflect real-world dynamic conditions. For example, an experiment with 187 participants on the effects of information and media literacy training on determining information authenticity found it only marginally effective [26].

Reactive methods focused on post-exposure interventions, such as flagging [61], source rating [40], and fact-checking [38, 56], have yielded interesting but inconsistent results. However, when combined with other mitigation strategies, these show potential. The limited studies on the effectiveness of flagging user-generated content indicate that flagging could deter negative behavior [60] or not [27] and did not impact truth judgments or beliefs but led recipients to take more time to evaluate the information [61]. Another study found that credible sources could reduce anxiety and enhance the effectiveness of combating fake news [65].

Fact-checking has proven more effective than citizen curation in combating misinformation [68], especially when strong social ties exist [56]. Coherently presenting information is more effective than fact-checking or source credibility in reducing fake news [86]. Furthermore, combining source ratings and user presentations to suppress fake news may encourage users to think more critically about the veracity of the content they encounter [39, 40].

### ***Correcting Fake News***

Correction messages have emerged as a popular strategy for countering the spread of fake news. Government agencies, fact-checkers, social media platforms, and individuals have embraced this approach as a proactive measure against misinformation. Our literature review suggests message framing is more effective than content in reducing fake news believability [52]. Corrections aim to refute or rebut information, often by presenting counterarguments or contradicting facts, evidence, or arguments. Frequently initiated by independent fact-checkers and government agencies, corrections are disseminated by regular social media users who repost and share them. The U.S. government created rumor control pages to counter rumors during crises like hurricanes, wars, and pandemics [87]. The Federal Emergency Management Agency and other government agencies actively combat online misinformation through dedicated rumor control pages.<sup>1</sup> However, as the Department of Homeland Security (DHS) observed, efforts to correct false information sometimes inadvertently amplify it [17], diverting crucial resources from disaster recovery efforts and hindering the agency's ability to assist those in real need.

Studies on the effectiveness of corrections in combating fake news have yielded mixed results. Corrections are primarily used as a mitigation technique [31], often through human intervention. Some studies noted that corrections could be effective in reducing fake news [88] because they present pertinent information to receivers [77], thereby undermining the credibility of false content [37]. In this study, we posit that when people are exposed to new information through correction updates, they revise their beliefs and create fresh perspectives [5], thus re-energizing the corrections. Studies on effective corrections typically focus on presenting corrections. Swire-Thompson et al. [77] investigated how modifying the presentation format of corrections affects fake news dependence; they found correction formats varying according to the order in which the constituent parts were presented, namely, "myth-first, fact-first, fact-only, and myth-only." Thus, while the impact on beliefs and reasoning depends on the format, providing corrective information is more important. In addition, combining true and false information in a single message may increase fake news awareness and curb its propagation [10]. Pairing correction messages with fake news and warning labels reduces the spread of rumor [66], a common approach often used by rumor mills when refuting fake news, as illustrated in Figure 1.2 in the Online Supplemental Appendix 1. Incorporating facts into correction messages influences engagement and exposes users to fake news [10]. However, repeating fake news adversely affects correction processes [89] because users may be unable to differentiate facts from myths. By contrast, evidence exists against the familiarity backfire effect when corrections that repeated fake news exposed users to novel fake news [23]. Indeed, contrary to previous research [17, 45], recent work shows that corrections have been largely ineffective, as they fail to reduce the spread of fake news and amplify it by creating awareness and engagement [45].

Our literature review reveals the following gaps:

- (1) Government agencies and fact-checkers refute fake news promptly; however, their approaches have generally been ineffective [45], making it necessary to investigate what constitutes effective corrections and how to tailor them.
- (2) While individual-level interventions could play a role in combating fake news, their effectiveness has been primarily explored in controlled laboratory settings, with minimal exploration of their real-world impact, owing to the complexities of tracing information diffusion dynamics [10, 37], necessitating research closer to real-world scenarios.
- (3) Prior work on the effectiveness of corrections has treated them as homogeneous [45], which can cause omitted variable bias. Identifying correction subpopulations or types can help us better understand the efficacy of correction messages.
- (4) A static approach (e.g., [43, 74]) limits user reactions to updates and restricts beliefs and responses to new information [5]. We propose using information updates to model a dynamic communication process.

Given the lack of an empirical solution based in theory to improve the efficacy of correction messages in mitigating the spread of misinformation, this study, grounded in the IMT framework and an empirical analysis of a novel panel dataset, examines different correction types, updates, and the moderating effects of including media files within the correction process to illuminate the design and influence of social media users' reactivity in countering fake news.

## Different Types of Correction Messages

In light of the mixed and inconclusive evidence regarding the effectiveness of corrections, their optimal use is debatable. King et al. [45] described the possible effect of the interaction between fake news and corrections, showing the conditions under which fake news stops propagating. They reported three different effects of corrections: (a) effective corrections reduce fake news propagation, (b) ineffective corrections amplify fake news propagation, and (c) there is no effect as corrections and fake news coexist or both decline over time due to a lack of interest. Hence, we consider corrections to be effective if they reduce fake news and ineffective if they increase fake news, yielding the following hypothesis:

*Hypothesis 1 (H1): There are different correction types with different effects on fake news.*

Rather than all correction messages having the same effect on fake news, we test whether different correction subpopulations have differentiated effects and find at least two subpopulations. In principle, all corrections are intended to reduce the spread of fake news, and separating them into subpopulations allows for differences in signs and magnitude. The empirical section formally tests for a differentiated effect and the signs of the effects.

## Information Manipulation Theory and Information Updates

To use IMT as a theoretical lens to examine information updates, we must align its central tenets to capture the potentially differentiated effects of corrections on fake news count.

While IMT has often been investigated within deceptive discourse as “manipulation,” we use the term “update” to describe the changes and modifications made to original correction messages during the resharing process. Information is often updated as it propagates [43]. For example, Shin et al. [74] observed that false rumors tend to return with notable content differences. Friggeri et al. [30] found that false textual memes were frequently updated as variants even after debunking, albeit in small volumes. As new information becomes available, people likely adjust their judgment after learning that a crucial piece of information upon which they based their opinion was inaccurate. While cognition may be responsible for the degree of updating beliefs [16], if a person is influenced by inaccurate information, this influence cannot simply be undone by pointing out the inaccuracy, and new or updated information that counters the effect must be presented. While it is unclear to what degree and how these updates affect the diffusion of fake news, by applying IMT, we can better understand how correction messages, especially their updates, affect the spread of fake news.

IMT, grounded in Grice’s theory of conversational implicature [35], posits that information can be manipulated or updated and involves elements that fall between truth and falsehood [58, 59, 90]. In general, the information within deceptive messages, for instance, varies systematically, regardless of the infinite dimensions of deception in interpersonal conversations. According to IMT, it involves modifying information along four dimensions: *quality*, *quantity*, *relation*, and *manner* [58, 59]. This variation includes the degree of important information disclosure, relevance, distortion, and conventional ambiguities [35]. Violations of these dimensions often occur through extremely subtle shading [91] and, similar to lying, it involves two primary ways: concealment and falsification. The former occurs when information is withheld and does not involve untruthfulness, and the latter occurs as an additional step that presents false information as truth and withholds the information. Concealment and falsification may often be necessary for deception to be effective, but sometimes, simply concealing information is enough [23]. Context-specific examples using violations and updates of these dimensions from our sample are presented next.

### **Quality, Quantity, Relation, and Manner Updates**

*Quality* is perhaps the most critical dimension in prototypical updated messages [59] because of “expectations regarding the veracity of the information presented” [58]. It is not a reflection of deception in the conventional sense but of “authenticity,” which is how much an individual (sharer of the information) references self. “Quality,” as we use the term, means sharing information without self-regulation, filtering, and employing self-referents [92], as posited by IMT. *Quality* violations involve using “fewer self-references” and references, similar to lying [63]. While attempting to alter the receiver’s perception of the message, they unconsciously distance themselves from the misrepresentation. Consider the following retweeted fake news message from our sample: “*I hear Gates is planning to microchip people with the vaccine or use it to reduce the world’s population.*” In this case, it is classified as misrepresentation because it involves altering how the message is perceived and does not include a reference. Propagators of falsehoods may append their statements with “I heard/hear” or “I read somewhere” to alter how the message is perceived, yet unconsciously distancing themselves from the fabricated message and decreasing its *quality* score. The fake

news example also includes a conjunction, “or,” indicating a lack of confidence in the statement’s veracity. It highlights the problematic use of “or.” It presents two distinct and unrelated conspiracy theories as if they were potentially interchangeable or even related outcomes. The “or” in this context suggests a lack of clear understanding or deliberate conflation of separate, baseless claims. It is not a “or not” case but rather “or a completely different, equally unsubstantiated claim.”

By contrast, consider the following correction message: *“It has come to our notice that some far right conspiracy theorists among (...) base, including those protesting state stay-at-home policies, falsely claim Gates is developing a vaccine that contains a microchip to track those who get injected. For more information, see the link below.”* This correction references the originators of the falsehood, redacted as (...), and the author(s) confidently claim ownership of the post using the phrase “our notice.” It also provides a website link (reference) to support the assertion because a correction’s credibility is contingent on how it manifests specific empirical referents. It concomitantly conveys the confidence to the recipient that these referents are verifiable. Newman et al. [63] argued that providing a source to support an assertion is also a *quality* measure. For that reason, this correction statement is more likely to be believed than the following: *“Someone told me that some far right conspiracy theorists, including those protesting state stay-at-home policies, falsely claim Gates is developing a vaccine that contains a microchip to track those who get injected.”* An increase in *quality* updates would be manifested in an increase in the degree of authenticity (and a decrease in filters) for every update in a correction.

We argue that, as updates of the original correction message exude authenticity, it may positively affect believability and draw the following hypothesis:

*Hypothesis 2 (H2): An increase in ‘quality’ updates increases the probability that the correction message is effective.*

The *quantity* dimension is among the most common update types because it represents an increase or decrease in the information. However, these violations are difficult to detect unless someone is privy to the truthful information being modified because they focus on simply altering the amount of information presented [90]. Often, there is no fabrication or distortion of information in these messages compared to the original message, nor do they have to deviate from their veracity. Consider the following example of fake news: *“Bill Gates will launch human-implantable capsules that have ‘digital certificates’ which can show who has been tested for the coronavirus and who has been vaccinated against it.”* Given the established expectations for disclosure, the statement above is relevant and appropriate to the original retweeted fake news, although it contains more words/information. *Quantity* violations involve “manipulating” the amount of information presented by using word omissions or inclusions. The following two fake messages: *“Bill Gates said he plans to track people with microchip implants so we can ‘fight the CoronaVirus,’”* and *“Bill Gates planning to use microchip implants to fight coronavirus,”* are relevant and involve no notable distortions to the original fake news. However, one involves inclusion, and the other, omission.

When receivers update the original corrections with more words, more information may be disseminated, and users may respond positively. Consider the following correction messages in response to the retweeted fake news: *“Please don’t believe the story of @BillGates using a corona vaccine to implant microchips in citizens! It is false and designed to discourage you from being vaccinated,”* and *“Bill Gates did not say he plans to track people*

*with the vaccine.*” The first includes more information on the statement being falsified, a negation and its veracity, while the second does not and includes a simple negation. However, neither distorts the original fake nor the correction message. The additional information is likely to encourage users to re-evaluate the fake news. However, omitting pertinent information can cause people to question the validity and credibility of the corrections (e.g., the second correction message). Furthermore, based on the elaboration likelihood model (ELM) of persuasion, the inclusion of information influences users to engage the central route to process information as opposed to relying on the peripheral route for processing the information [93], which overcomes the barriers associated with lazy thinking, improving believability. Thus, an increase in the *quantity* updates is manifested as an increase in the number of words per update in a correction that brings back previously withheld facts, yielding the hypothesis:

*Hypothesis 3 (H3): An increase in ‘quantity’ updates increases the probability that the correction message is effective.*

Another linguistic alternative is to update information regarding its relevance. *Relation* updates occur when a proper response is not provided to an original message but is diverted from the main subject [58]. Consequently, they frequently initiate new topics that deviate from the topic of discussion and are “irrelevant.” Even though *quantity* updates may not always be relevant, updates of *relation* are most likely a result of extreme *quantity* violations [58]. As the *quantity* update becomes very high or low through inclusion or exclusion, respectively, they will likely become irrelevant compared to the original. Hence, the variables capturing *relation* and *quantity* updates may be more correlated than the others. We argue that diverting narratives may positively affect believability by providing alternative perspectives. For instance, consider the fake news story: “Please scan your COVID19/ID2020 vaccination microchip to apply for unemployment. If you do not have a chip yet . . .,” and the resulting correction update responses: “I am a factual person. Really? A factual person with an associate’s degree in science being misinformed on the basics of how viruses spread? You have cited no studies/evidence of your claims of microchipping or the United States creating the coronavirus. It’s complete nonsense,” and “You can not scan your COVID19/ID2020 vaccination microchip to apply for unemployment.” The first counterargument deflects by focusing on the sender’s qualifications and educational status, thus preventing the familiarity effect [67, 94], as opposed to the second correction update, which simply repeats the fake news but with a negation. It has long been argued that corrections that repeat the falsehoods are less effective and increase the likelihood of reinforcing its reach to new audiences [76].

An increase in *relation* updates manifests as an increase in the deflections used per update in a correction that deviates from the fake news story. Therefore, draw the following hypothesis:

*Hypothesis 4 (H4): An increase in ‘relation’ updates increases the probability that the correction message is effective.*

The *manner* dimension is one of clarity of expression, owing to the level of detail provided (i.e., the statement’s quality of being easily understood depends on the number of specifics included) or ambiguity [6], described as being open to more than one interpretation. Falsehoods rely on ambiguity, subjective beliefs, and experiences

influenced by personal perspectives, whereas corrections rely on facts and verifiable information not influenced by subjective beliefs. In this study, because we are interested in corrections' effectiveness, not falsehoods, we focus instead on the level of detail or specificity relative to the original correction and not vagueness. Specificity is determined by the way a sentence "relates uniquely" to a particular topic [86] and can often be adjusted to reflect the disseminator's intentions [34]. Clear statements that provide readers with an appropriate level of detail are often considered effective. For example, the fake news statement, "*That's how they're planting the microchip guys,*" is quite low in specificity and is open to multiple interpretations. As a result, specificity is critical for reading comprehension [18] and establishing shared interest in discourse [95]. Empirical statements (e.g., corrections) that vary widely in specificity can have important implications for credibility [71]. A correction message's lack of specific empirical content can diminish its persuasive effect on social media platforms, where the audience is heterogeneous but brought together by common interests. If a low-specificity message appears effective, it is usually the result of the receiver projecting their own experiences onto the ambiguous empirical references. However, a message that is too elaborate can also cause information overload.

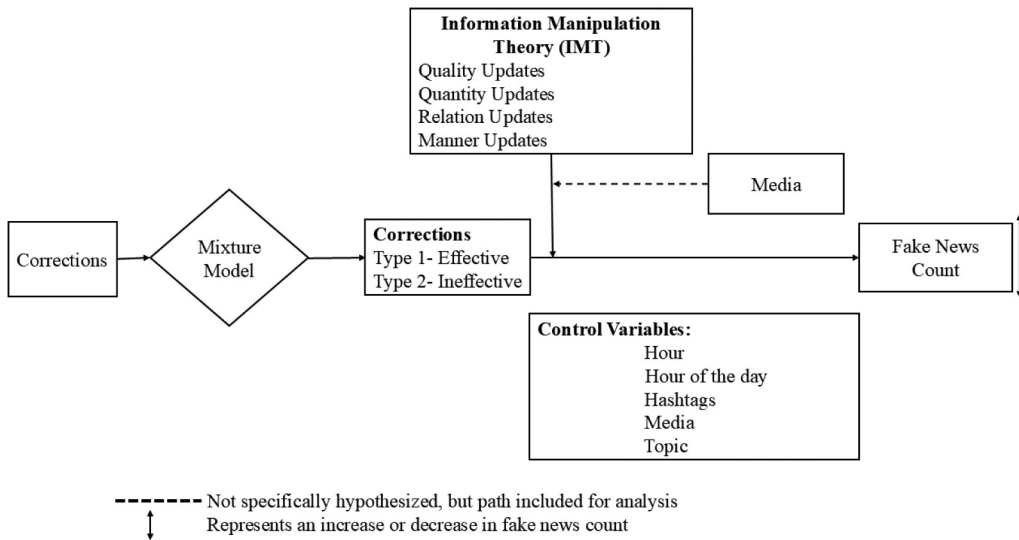
Although H3 implies that more information increases the effectiveness of corrections, we contend that the level of detail provided in a correction message can affect its efficacy. The information provides the framework for presenting details, which enhances the information's depth, richness, and complexity. In essence, details are the building blocks of information that add nuance and complexity to the overall picture. IMT posits that being specific positively affects a message's persuasiveness, but one can argue that the marginal utility of each additional detail relative to the correction may even turn negative. We argue that effective corrections contain facts and an objective reality and that each additional detail dilutes this objectivity.

Consider the correction message: "*The quantum dot dye technology is not a microchip or human-implantable capsule and there are no plans to use this for coronavirus.*" This one may be more easily believed than the message: "*The tiny nanocrystals with extraordinary optical and electrical properties consisting of four copper ions dye technology is not an integrated circuit (IC) or human-implantable capsule and there are no plans to use this for coronavirus.*" We argue that the first correction contains just the right amount of detail and is more easily believed. Although informative and relevant, the second option includes excessive detail, which may encourage subjective interpretations.

Furthermore, pathological liars, those with a condition known as pseudologia fantastica, often provide elaborate or too many details [41, 49]. In general, overly vivid, precise messages with too much detail or specificity can be less persuasive and cause doubt [14, 25, 29]. An increase in *manner* updates increases the degree of detail per correction update, potentially affecting the objectivity of the presented facts, leading to our final hypothesis.

*Hypothesis 5 (H5): An increase in "manner" updates reduces the probability that the correction message is effective.*

Overall, we posit that *quality*, *quantity*, *relation*, and *manner* updates moderate the effects of different correction message types on fake news count. The conceptual model is presented in Figure 1.



**Figure 1.** The conceptual model.

## Data and Empirical Model

Our data from Twitter (now called X) comprised verified fake messages and the corresponding correction messages during the first three months of the COVID-19 pandemic. The coronavirus pandemic was first reported in 2019 in Wuhan, China, and has reached every country worldwide. As of October 2024, the World Health Organization estimated 776 million cases and 7 million deaths. Fake news propagates during significant events when information received about those events is subjectively ambiguous, lacking, or both [2]. However, people update their beliefs as events progress and the public gathers more information [5].

The data collection steps were as follows. The first step was to identify and collect, in real-time, all tweets from January 1 to March 31, 2020, using the Twitter stream API. Twitter was chosen because it was one of the most commonly used social media platforms for news and information [96]. We identified 76 keywords and hashtags related to the pandemic while updating this list, such as #Coronavirus, #Koronavirus, #Corona, #Wuhancoronavirus, #Kungflu, #sars-cov- 2, #covid-19, #coronavirus, and #COVID\_\_19 iteratively. This resulted in 83,998,658 tweets and retweets (11,579,448 in January, 32,725,042 in February, and 39,694,168 in March). Accurate identification is key to countering misinformation on social media [70]. Using the Federal Emergency Management Agency (FEMA) and five fact-checking organization websites, we separately identified 25 fake news stories and their associated keywords using either FEMA's rumor control website and their Twitter timelines or from verified fake stories that appeared in at least four of these five independent fact-checking websites: Snopes, Factcheck.org, PolitiFact, Truth or Fiction, and BS Detector. Fact-checking websites ensure the selected topics' veracity and accuracy [83], allowing us to label these 25 topics as verified fake news. Online Supplemental Appendix 1 Tables 1.1 and 1.2 list the topics and keywords.

Using the identified keywords and following King et al. [45], keyword searches were performed on the Twitter data using one or more keyword combinations to extract all possible fake news messages related to each news story. The original keywords were directly extracted from FEMA and fact-checking websites and augmented with a Latent Dirichlet Allocation (LDA) analysis [9]. LDA identifies “underlying topics and their similarities.” We leveraged the SpaCy, NLTK, and GenSim libraries for our LDA analysis. After preprocessing (i.e., removing stop words and lemmatization), we utilized a grid search to determine the most statistically appropriate number of topics. Although the coherence and perplexity values (and the weights of the keywords) were insightful, graduate students qualitatively reviewed the resulting subtopics and keywords associated with each topic to select new keywords to augment our search. After deduplicating the final data using their unique IDs to prevent redundancy, the new keywords accounted for 3.39 percent (41,544) of the newly identified messages, suggesting that approximately 3 percent of the reshapes were from the updated keywords. While we had 25 distinct topics of interest, there were several shared common keywords; hence, a qualitative review of selected topics related to each keyword was appropriate. Keywords centered on who, what, how, and where. Our keyword extraction steps are presented in the Online Supplemental Appendix 1.

Because of the volume of data, a computational approach to categorize data would have been more practical and less time-consuming [54], but would also generate higher levels of error because categorizing fake news, corrections, and noise can be complex [42]. Manual data classification is common for medium and large datasets; for example, Shin et al. [74] manually classified the veracity of 439,556 rumors on Twitter, whereas King et al. [45] hand-coded approximately 250,000 tweets. Thus, we opted for accuracy at the expense of speed. We “manually single-coded” each tweet per topic as fake news or a correction message with the help of 18 graduate students.<sup>2</sup> The dataset was refined by employing filtering techniques, sorting similar content, and using negation to identify contradictory information. After initial filtering with the keywords, 2,682,001 of the 83,998,658 observations remained (January with 188,559; February with 867,676; and March with 1,625,766). We asked our coders to determine (1) whether the tweets were about the original fake news and (2) whether the attitude of the tweet was supported, rejected, or unclear. Tweets debunking or rebutting fake news were coded as corrections, whereas those repeating, citing, or corroborating verified fake news were coded as fake news. Two coders were assigned to manually assess the accuracy of the 25 topics using a 1 percent random sample of the data. When there was a disagreement between the coders, we either removed the item or assigned the correct label.

Of the 25 topics, we excluded two with relatively few corrected observations. In addition to extracting the content from each tweet, we obtained user metadata related to the post, snapshots of the author’s profile, and other related media content. Following King and Wang [44], we extracted several content and linguistic features. We tracked the hourly propagation of fake news and corrections for each topic to determine the propagation count. Our approach resulted in 1,221,887 observations and 38,617 topic-hour data points representing individual data points associated with specific topics across different hours. Table 1 presents the summary statistics, and Table 2 lists the variables and their definitions.

A list of topics is presented in the Online Supplemental Appendix 1 Table 1.1. The Variance Inflation Factor (VIF; all) was 4.63, while the VIF of the baseline model was 1.00, based on which no multicollinearity problem was anticipated [81].

**Table 1.** Summary statistics.

|                                     | (1)   | (2)   | (3) | (4)    |
|-------------------------------------|-------|-------|-----|--------|
| Variables                           | mean  | SD    | min | max    |
| <i>FakeNews<sub>it</sub></i>        | 34.37 | 124.2 | 0   | 3,854  |
| <i>Corrections<sub>it</sub></i>     | 5.809 | 23.90 | 0   | 1,045  |
| <i>Media<sub>it</sub></i>           | 0.354 | 2.50  | 0   | 202    |
| <i>Hash<sub>it</sub></i>            | 1.118 | 5.736 | 0   | 386    |
| <i>Quality<sub>it</sub></i>         | 87.81 | 884.0 | 0   | 79,881 |
| <i>Quantity<sub>it</sub></i>        | 85.26 | 917.9 | 0   | 96,766 |
| <i>Relation<sub>it</sub></i>        | 2.661 | 25.13 | 0   | 2,502  |
| <i>Manner<sub>it</sub></i>          | 2.053 | 20.81 | 0   | 2,056  |
| <i>Hours<sub>it</sub></i>           | 839   | 484.7 | 0   | 1,678  |
| <i>q<sub>it,1</sub><sup>a</sup></i> | 0.542 | 0.394 | 0   | 1      |

Notes: The sample size is 38,617 for the probability of having a Type-1 (effective) message, <sup>a</sup> based on Model (2) of Table 4. Variable descriptions are presented in Table 2, and vary by topic *i* and hour *t*.

**Table 2.** Variable definitions.

| Variable                       | Definition                                                                                                 |
|--------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Dependent variable</b>      |                                                                                                            |
| <i>FakeNews<sub>it</sub></i>   | Count of fake news                                                                                         |
| <b>Independent variables</b>   |                                                                                                            |
| <i>Correction<sub>it</sub></i> | Count of correction messages                                                                               |
| <i>Media<sub>it</sub></i>      | Count of media files in a correction message                                                               |
| <i>Hash<sub>it</sub></i>       | Count of hashtags in a correction message                                                                  |
| <i>Quality<sub>it</sub></i>    | The difference in the authenticity between the original correction and subsequent messages                 |
| <i>Quantity<sub>it</sub></i>   | The difference in the mean word count per sentence between the original correction and subsequent messages |
| <i>Relation<sub>it</sub></i>   | The difference in the cosine similarity between the original and subsequent correction messages            |
| <i>Manner<sub>it</sub></i>     | The difference in the specificity of an original correction and subsequent correction messages             |
| <i>Hours<sub>it</sub></i>      | Hour the tweet was posted                                                                                  |

### Operationalization of Variables

Our dependent variable is the aggregate count of fake news per topic over time, whereas corrections, our main independent variable, is captured as the aggregate count of correction tweets per topic over time. We measured *quality* updates as the difference between the authenticity scores of the original and subsequent corrective tweets [57, 63, 97]. Higher values indicated higher levels of authenticity. We measured *quantity* updates using the difference between the mean word counts per sentence of the original and subsequent correction messages [80]. *Relation* violations were captured as the cosine distance between the original and subsequent correction tweets. The closer the subsequent tweets are to the original correction, the lower the *relation* violation. Cosine distance is the inverse of cosine similarity (1 - cosine similarity), a technique for calculating the similarity between textual content using an agglomerative approach [33]. As the cosine distance approaches zero, the correction messages become increasingly similar (and relevant) to the original correction news. The cosine similarity between vectors *X* and *Y* is denoted as  $\text{cosine}(X; Y) = X \cdot Y / (||X|| * ||Y||)$ , where  $||X||$  and  $||Y||$  are the Euclidean norms for *X* and *Y*, respectively. Higher values indicated a measure of irrelevance. Following Ko et al. [48], we measured manner updates, a self-assembling architecture [98] that compares the specificity of the original correction and

subsequent correction messages. Based on the degree of detail in a sentence, this scale ranged from 0 (most general) to 1 (most detailed). Higher values indicate greater specificity.

### Empirical Model Specification

This section presents the development of an empirical model for assessing the effects of correction messages on the diffusion of verified fake news. We also test for the existence of a nonlinear effect, as different correction message types are “allowed” to have differentiated effects on fake news. Our modeling approach addresses the potential omitted variable problem, as corrections can remain unobserved. Prior research has often treated the effects of correction messages as homogeneous; however, diverse individuals in varying contexts of uncertainty and information *quality* respond differently to correction messages. For example, populations are often divided into sub-populations, and the differences between these subpopulations may be unobserved. Our approach to this problem uses mixtures [82], an unsupervised probabilistic model, to represent the presence of subpopulations within an overall population. We characterized the probability of belonging to a particular subpopulation or type and estimated the distinct parameters of a regression model. This allowed us to classify messages and infer the behavior of each subpopulation. In a mixture model, observations arise from various probability distributions and are increasingly favored over heuristic methods because of their sound mathematical foundation and intuitive interpretation [99]. In addition, overlapping clusters corresponding to the mixture model components can be identified using a probabilistic clustering approach. Several studies rely on mixed models to detect unobserved relationships. Kumar et al. [50] identified spammers on Yelp using mixture models to derive distributions to define reviewer behavior; their approach outperformed previous unsupervised anomaly detection algorithms in identifying latent critical features. Escobari and Hernandez [24] leveraged a mixture model to identify and assess unobserved consumer types. We used mixtures to explicitly account for the limited information on potentially different types of corrections.

The equations that characterize fake news are as follows:

$$\log (FakeNews)_{it} = \begin{cases} \alpha_1 + \beta_1 \log (Corrections)_{it} + X_{it}\delta_1 + \mu_{i,1} + \eta_{t,1} + \varepsilon_{it,1} & \text{if } \theta = 1 \\ \alpha_2 + \beta_2 \log (Corrections)_{it} + X_{it}\delta_2 + \mu_{i,2} + \eta_{t,2} + \varepsilon_{it,2} & \text{if } \theta = 2 \\ \vdots & \vdots \\ \alpha_N + \beta_N \log (Corrections)_{it} + X_{it}\delta_N + \mu_{i,N} + \eta_{t,N} + \varepsilon_{it,N} & \text{if } \theta = N \end{cases} \quad (1)$$

where  $\log (FakeNews)_{it}$  and  $\log (Corrections)_{it}$  are the logarithms of the counts of fake news and correction messages for topic  $i$  during hour  $t$ . The set of observable explanatory variables that serve as controls is contained in the matrix  $X_{it}$ .  $\mu_i$  are topic-specific fixed effects that control for time-invariant observed and unobserved differences across topics, while  $\eta_t$  are hour-of-the-day fixed effects accounting for topic-invariant observed and unobserved differences across different times of the day.  $\varepsilon_{it}$  represents the error term. Equation (1) models the existence of  $N$  different message types  $\theta$ , with  $\theta = 1, 2, \dots, N$ . The constant terms absorb all factors associated with unobserved correction message types. We are interested in estimating  $\beta_\theta$ , the differentiated effects of correction messages on fake news. Note that the vector of coefficients on  $X_{it}$  and both sets of fixed effects can vary by

**Table 3.** Determining the number of types.

| Number of Types (N) | SBIC  | Log Likelihood | Likelihood Ratio |
|---------------------|-------|----------------|------------------|
| 1                   | 3.848 | -74,167        |                  |
| 2                   | 3.219 | -61,860        | -24,614          |
| 3                   | 3.390 | -65,026        | 6,331            |

Note:  $N=38,617$ .

type. If we assume that the error terms in Equation (1) follow a normal distribution such that  $\varepsilon_{it,\theta} \sim N(0, \sigma_{\varepsilon,\theta}^2)$  for  $\theta = 1, 2, \dots, N$ , the log-likelihood of the  $k$ th (topic-hour) is given by:

$$\log l_k = \log \left[ \sum_{\theta=1}^N \frac{q_{\theta}}{\sigma_{\varepsilon,\theta} \sqrt{2\pi}} \exp \left( -\frac{\varepsilon_{k,\theta}^2}{2\sigma_{\varepsilon,\theta}^2} \right) \right] \quad (2)$$

where  $q_{\theta}$  is the mixing parameter defined as the probability of being in a regime of type  $\theta$  messages and  $\sum_{\theta=1}^N q_{\theta} = 1$ . Then, Equation (2) shows us that each  $k$ th observation can be linked with a specific message type  $\theta$  for  $\theta = 1, 2, \dots, N$ . An additional interpretation is that message types coexist during each observed hour and where the type-mixture is time-varying as captured by  $q_{\theta}$ .

As the specification in Equation (1) allows the coefficients to change across types, it is of interest to model the probability of observing a given message type. The probability of observing a type  $\theta$  message is modeled as

$$q_{it,\theta} = \frac{\exp(\lambda_{\theta} G_{it})}{1 + \sum_{s=1}^{N-1} \exp(\lambda_s G_{it})} \quad (3)$$

where  $G_{ij}$  is the set of observable message characteristics. Equation (3) can be used to associate the identified message type with characteristics such as information updates.

### Model Selection

Correction messages are designed to counteract fake news but may be ineffective and cause blowback [52]. King et al. [45] depicted the possible effects of the interaction between fake news and corrections, including scenarios in which fake news could eventually stop spreading or be amplified.

The first step in our estimation involved determining whether allowing for different types represents an improvement in model fit compared to pooling all messages together. Although Eq. (1) allows for multiple types, no statistically significant difference may exist between them [45]. Table 3 reports the maximized value of the log-likelihood function, the Schwarz Bayesian Information Criterion (SBIC), and the corresponding Likelihood Ratio test, comparing the fitness between the pooled model and the two- and three-type models. The Bayesian approach provides the odds of one model versus another based on an estimate derived from data [79]. The log-likelihood and SBIC favor selecting the two-type model over the pooled model and the model with three types.

Moreover, we reject the null hypothesis that the observed data support the single-type model, as the LR of -24,614 is associated with a p-value of 0.000. Overall, it provides strong

evidence supporting the existence of two types of correction messages. Each observed correction message can be linked to Type 1 or Type 2 with a specific probability.

## Empirical Results

### *Pooled Types*

The first column in Table 4 shows the maximum likelihood estimation (MLE) results when we assume all correction messages have the same effect on fake news. When pooled across all topic-hour observations, the coefficient of  $\log(\text{Corrections})_{it}$  shows that a 1 percent increase in correction messages increases the number of fake news messages by 0.11 percent, with a statistically significant coefficient. This positive effect implies that corrections, rather than mitigating the spread of fake news, strengthen them by drawing attention to misinformation. It is consistent with recent findings that corrections amplify fake news by increasing awareness of and possibly strengthening prior beliefs regarding it [45]. The negative and statistically significant coefficient of  $\text{Hours}_{it}$  in Model (1) for the fake news equation indicates that the number of fake news decays over time. On average, fake news messages that are 1,000 hours (about 41.6 days) older have about 10.1 percent fewer message counts than newer messages.

### *Differentiated Effectiveness of Correction Messages*

The first set of estimates from the joint estimation of Eq. (1) and Eq. (3) is presented in Model (2) and Table 4. When allowing correction messages to have a differential effect on fake news, we tested whether the difference between these marginal effects was statistically significant. The bottom of Table 4 presents the statistic and the corresponding p-value to test the null hypothesis that  $\beta_1 = \beta_2$ . We reject the null hypothesis, providing strong empirical evidence that different types of correction messages have varying effects on fake news.

Model (2) shows two distinctive correction message types when examining the signs and magnitudes of the coefficients. The most crucial difference is that Type-1 messages have a negative effect on fake news count, whereas Type-2 messages have a positive effect. Our mixture model of the unobserved types can separate correction messages depending on how they affect fake news. Empirical evidence suggests that Type-1 correction messages effectively deter social media users from sharing fake news. The coefficient is statistically significant, showing that a 1 percent increase in Type-1 corrections decreased fake news by 0.896 percent. By contrast, the model identifies Type-2 messages that positively affect the spread of fake news. A 1 percent increase in Type-2 messages increased the number of fake news items by 1.37 percent, with the effect being statistically significant.

The third model in Table 4 augments  $G_{it}$  in Eq. (3) to characterize the probability of correction messages being Type-1 effective depending on  $\text{Hours}_{it}$ . A negative and statistically significant coefficient indicates that the likelihood of encountering such messages decreases over time. The point estimate indicates that the probability decreased by approximately 0.917 percent after two days.<sup>3</sup> One possible explanation could be the level of fatigue, as individuals are less likely to be exposed to effective correction messages over time. The paths of Type-1 and Type-2 messages are illustrated in Figure 2. Their classification follows Model (3) and Table 4. The shaded areas represent the 95 percent confidence bands and

**Table 4.** Mixture model estimates.

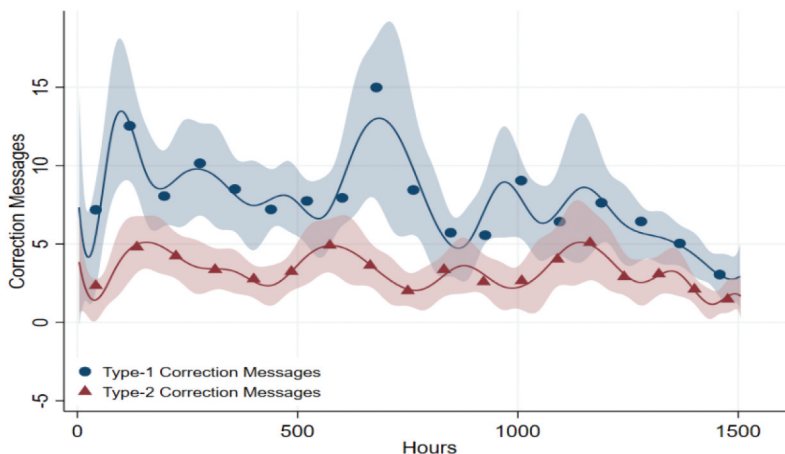
| Model:                                                           | (1)                   | (2)                   |                       | (3)                   |                       |
|------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Type $\theta$ :                                                  | Pooled                | Type-1                | Type-2                | Type-1                | Type-2                |
| Fake News Equations:                                             |                       |                       |                       |                       |                       |
| $\log(\text{Corrections})_{it}$                                  | 0.106***<br>(0.0129)  | -0.896***<br>(0.0241) | 1.370***<br>(0.00623) | -0.897***<br>(0.0235) | 1.371***<br>(0.00620) |
| $\text{Hours}_{it}/1000$                                         | -0.101***<br>(0.0174) | -0.225***<br>(0.0238) | 0.0147**<br>(0.00732) | -0.199***<br>(0.0241) | 0.0149**<br>(0.00720) |
| Constant                                                         | 0.0286<br>(0.0412)    | 0.432***<br>(0.0560)  | -0.356***<br>(0.0166) | 0.433***<br>(0.0560)  | -0.355***<br>(0.0166) |
| $\sigma_\varepsilon^2$                                           | 2.727***<br>(0.0174)  | 2.346***<br>(0.0492)  | 0.139***<br>(0.00686) | 2.344***<br>(0.0480)  | 0.140***<br>(0.00665) |
| Probability of Type $\theta = 1$ Effective Corrections Equation: |                       |                       |                       |                       |                       |
| $\text{Hours}_{it}/1000$                                         |                       |                       |                       | -0.191***<br>(0.0272) |                       |
| Constant                                                         |                       | 0.172***<br>(0.0188)  |                       | 0.168***<br>(0.0184)  |                       |
| Mean of $\text{FakeNews}_{it}$                                   | 34.366                | 49.69                 | 21.23                 | 49.57                 | 21.36                 |
| Topic Fixed Effects                                              | Yes                   | Yes                   |                       | Yes                   |                       |
| Time Fixed Effects                                               | Yes                   | Yes                   |                       | Yes                   |                       |
| Observations                                                     | 38,617                | 38,617                |                       | 38,617                |                       |
| Log Likelihood                                                   | -74,168               | -61,861               |                       | -61,836               |                       |
| $\chi^2$ of $H_0: \beta_1 = \beta_2$                             |                       | 7,195                 |                       | 7,616                 |                       |
| P-value of $H_0: \beta_1 = \beta_2$                              |                       | 0                     |                       | 0                     |                       |

Notes: The dependent variable is  $\log(\text{FakeNews})_{it}$ . Heteroscedasticity robust standard errors in parentheses. \* significant at 10 percent; \*\* significant at 5 percent; \*\*\* significant at 1 percent.

were constructed using a studentized t-statistic. The figure was obtained using the nonlinear regression method described in Cattaneo et al. [11].

While topic and hour-of-the-day fixed effects help control for unobserved shocks common across topics or hours, additional variation might likely come from information obtained from other sources (e.g., TV, radio, or personal interaction). Additional specifications include different dimensions of correction updates.

Figure 2 illustrates that Type-1 effective messages have a higher count than Type-2 ineffective correction messages. Consistent with the larger magnitude of the point estimate of  $\beta_2$ , when compared to the point estimate of  $\beta_1$ , we argue that Type-2 are more potent



**Figure 2.** Evolution of effective and ineffective correction messages over time. Type-1 effective and Type-2 ineffective correction messages as they evolve ( $\text{Hours}_{it}$ ), based on Model (3) of Table 4.

than Type-1. This highlights the importance of frequency and potency in the diffusion process.

Using the model (2) estimates in Table 4, Figure 3 illustrates the probability that a correction message in our sample is Type-1 effective. This figure is consistent with well-defined types. There is a relatively large probability mass at zero, indicating that a large proportion of messages are classified as Type-2.

Message Information Updates

We now test our identification strategy by assessing the effect of information updates on the probability of Type-1 effective corrections. Figure 2 shows the importance of frequency and potency in the diffusion process. We argue that the different dimensions of information updates may explain the potency of different types of correction messages in reducing the spread of fake news. Previous research noted information updates during propagation [44, 74] but largely focused on their effects. Table 5 presents the specifications of the matrix  $G_{it}$  in Eq. (3) to characterize the probability of a Type-1 effective correction. Model (1) includes  $\log (Media)_{it}$  and  $\log (Hash)_{it}$  because hashtags serve to promote and propagate ideas and topics [13], and fact-checking supported by external content is considered credible [73]. The positive and statistically significant coefficient of the log of hashtags strongly supports the hypothesis that messages including hashtags are more likely to be effective (Type-1). This positive sign can be explained by hashtags that help users find effective correction messages. The negative and statistically significant coefficient of media in the probability of the Type-1 equation shows that as the higher the number of media files in correction messages increases, the less effective they are. Finally, the coefficient of hours is consistent with the results in Table 4.

Model (2) augments the Type-1 probability equation with  $Quality_{it}$ ,  $Quantity_{it}$ ,  $Relation_{it}$ , and  $Manner_{it}$  updates because IMT states that information is often updated within those four dimensions [58, 59].  $Quality_{it}$ ,  $Quantity_{it}$ , and  $Relation_{it}$  information updates have positive and statistically significant coefficients, which indicate that updates

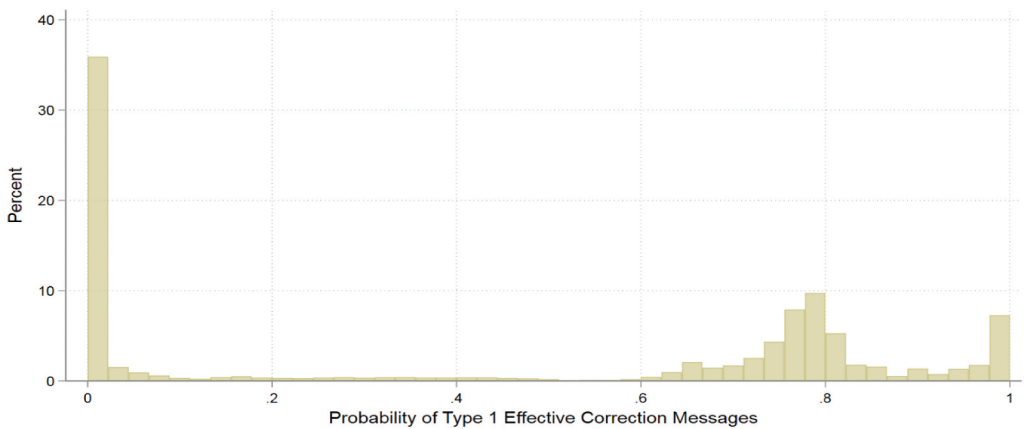


Figure 3. Probability of Type-1 effective correction messages, based on model (2) of Table 3.

**Table 5.** Probability of Type-1 effective collections: The role of information updates.

| Model:                                                           | (1)                  |                      | (2)                  |                      | (3)                  |                      |
|------------------------------------------------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| Type $\theta$ :                                                  | Type-1               | Type-2               | Type-1               | Type-2               | Type-1               | Type-2               |
| Fake News Equations:                                             |                      |                      |                      |                      |                      |                      |
| $\log(\text{Corrections})_{it}$                                  | -0.887***<br>(0.025) | 1.370***<br>(0.006)  | -0.863***<br>(0.029) | 1.367***<br>(0.006)  | -0.856***<br>(0.030) | 1.366***<br>(0.006)  |
| $\text{Hours}_{it}/1000$                                         | -0.198***<br>(0.024) | 0.015**<br>(0.007)   | -0.198***<br>(0.024) | 0.018**<br>(0.007)   | -0.199***<br>(0.024) | 0.019**<br>(0.007)   |
| Constant                                                         | 0.437***<br>(0.056)  | -0.355***<br>(0.017) | 0.443***<br>(0.056)  | -0.351***<br>(0.016) | 0.445***<br>(0.056)  | -0.352***<br>(0.016) |
| $\sigma^2_\varepsilon$                                           | 2.361***<br>(0.050)  | 0.137***<br>(0.007)  | 2.407***<br>(0.056)  | 0.129***<br>(0.007)  | 2.419***<br>(0.059)  | 0.127***<br>(0.008)  |
| Probability of Type $\theta = 1$ Effective Corrections Equation: |                      |                      |                      |                      |                      |                      |
| $\log(\text{Media})_{it}$                                        |                      | -0.170***<br>(0.042) |                      |                      |                      | -0.148***<br>(0.044) |
| $\log(\text{Hash})_{it}$                                         |                      | 0.171***<br>(0.029)  |                      |                      |                      | 0.139**<br>(0.029)   |
| $\text{Quality}_{it}$                                            |                      |                      | 0.313***<br>(0.037)  |                      |                      | 0.313***<br>(0.037)  |
| $\text{Quantity}_{it}$                                           |                      |                      | 0.116***<br>(0.036)  |                      |                      | 0.117***<br>(0.037)  |
| $\text{Relation}_{it}$                                           |                      |                      | 0.724***<br>(0.140)  |                      |                      | 0.711***<br>(0.139)  |
| $\text{Manner}_{it}$                                             |                      |                      | -1.391**<br>(0.157)  |                      |                      | -1.387***<br>(0.157) |
| $\text{Hours}_{it}/1000$                                         |                      | -0.179***<br>(0.027) |                      | -0.059**<br>(0.029)  |                      | -0.052*<br>(0.029)   |
| Constant                                                         |                      | 0.173***<br>(0.019)  |                      | 0.201***<br>(0.023)  |                      | 0.206***<br>(0.024)  |
| Mean of $\text{FakeNews}_{it}$                                   | 49.89                | 20.99                | 49.66                | 21.81                | 49.95                | 20.96                |
| Topic Fixed Effects                                              | Yes                  |                      | Yes                  |                      | Yes                  |                      |
| Time Fixed Effects                                               | Yes                  |                      | Yes                  |                      | Yes                  |                      |
| Observations                                                     | 38,617               |                      | 38,617               |                      | 38,617               |                      |
| Log Likelihood                                                   | -61,815              |                      | -61,552              |                      | -61,509              |                      |
| $\chi^2$ of $H_0: \beta_1 = \beta_2$                             | 6,613                |                      | 5,050                |                      | 4,533                |                      |
| P-value of $H_0: \beta_1 = \beta_2$                              | 0                    |                      | 0                    |                      | 0                    |                      |

Notes: The dependent variable is  $\log(\text{FakeNews})_{it}$ . Heteroscedasticity robust standard errors in parentheses. \* significant at 10 percent; \*\* significant at 5 percent; \*\*\* significant at 1 percent.

increase the probability of a correction message being effective. However,  $\text{Manner}_{it}$  has a negative and significant coefficient, indicating that the probability that a message is effective decreases as corrections are updated with more detail. All specifications include hour-of-day fixed effects. The coefficient of  $\text{Hours}_{it}$  is consistent with previous specifications. As time passes, correction messages are more likely to become ineffective. Thus, consistent with the literature, novel information is likely to be more effective earlier in the diffusion process and lose effectiveness over time. Model (3) shows that the results are robust to the inclusion of hashtags and media in the probability of the Type-1 equation. Figure 4 illustrates how the probability of receiving an effective message increases when combating fake news with *quantity* and *relation* information updates.

Online Supplemental Appendix 2 Table 2.1 presents additional estimates to test the hypotheses of the nonlinear effects of *quality*, *quantity*, *relation*, and *manner* of updates on the probability of receiving effective correction messages. Evaluated at its sample mean, Model (1) shows a positive but decreasing marginal effect of  $\text{Relation}_{it}$ . Furthermore, at the sample mean of  $\text{Manner}_{it}$ , Model (2) indicates a negative effect of  $\text{Manner}_{it}$  but with

a decreasing magnitude as *manner* increases. Model (3) also captures the nonlinear effect, where  $Quantity_{it}$  has a positive and increasing marginal effect on the probability of effective corrections. Finally, Model (4) considers the nonlinearities in  $Quality_{it}$ . In this specification, the coefficient of  $Quality_{it}^2$  is not statistically significant. Therefore,  $Quality_{it}$  has a positive and linear effect on the probability of having Type-1 effective correction messages. Overall, except for *quality* updates, there is strong, statistically significant evidence of nonlinearities in the effects of information updates.

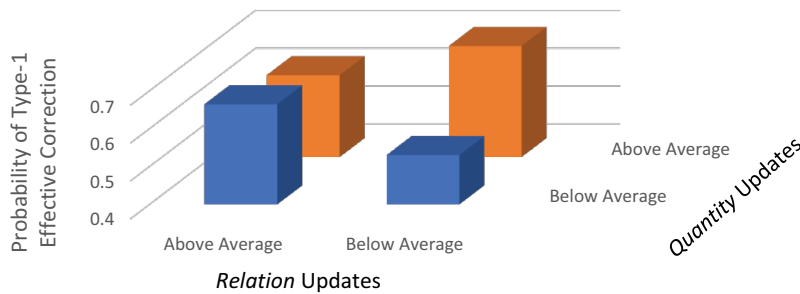
Online Supplemental Appendix 2 Table 2.3 explains the role of media files on the marginal effects of information updates on the probability of effective corrections. Prior research suggests that arguments are more credible if they provide more information to support an advocated position [73]. Model (3) shows that media files play no role in the effects of *manner* and *relation*; however, their interactions with *quality* and *quantity* are statistically significant. Consistent with Table 5, media files increase the positive effect of  $Quality_{it}$  and decrease that of  $Quantity_{it}$ . The results are consistent with the literature that updates may not always operate in isolation [58].

Additional Robustness Checks

Online Supplemental Appendix 2 Table 2.2 reports the natural robustness check to construct Eq. (1) and Eq. (3), lagging the variables on the right-hand side. The primary results reported in Table 5 hold with relatively small differences in the magnitudes of the coefficients. The only difference is that the coefficient of  $Hours_{it}$  in Models (2) and (3) is no longer statistically different from zero, which is not central to our contribution.

Fact-checking may select, highlight, or accentuate specific types of rumors. This may have created a selection bias in our sample of fact-checked messages. Online Supplemental Appendix 1 Figure 1.1 presents the coefficient plots and confidence intervals that illustrate wide heterogeneity in corrections and fake news, proving that our sample is unlikely to suffer from a selection bias. Table 1.1 in the Online Supplemental Appendix 1 presents a list of topics. Following Davis et al. [15] and Vosoughi et al. [83], the number of observations in each topic controls for the influence of nonhuman entities (e.g., bots). These entities are preprogrammed and perform actions, such as replying or resharing, which may influence message types and their effects on fake news.

The fourth robustness check includes a post-estimation analysis using Twitter data collected during four other crisis events: Hurricane Harvey in 2017, Ebola pandemic in 2014, Nepal Earthquake in 2015, and Fort McMurray Wildfire in 2016. Graduate students selected the crisis events based on their severity and impact. The descriptive statistics and



**Figure 4.** Probability of Type-1 effective correction. The probability of type-1 effective correction messages is based on Model (3) of Table 5. The above-average quantity and relation updates refer to a large variation between the original and corrective fake news messages.

estimation results are provided in the Online Supplemental Appendix. They provide strong empirical evidence that multiple correction types exist across all four additional datasets, showing further support for H1. This robustness check provides preliminary evidence for generalizability beyond our main sample.

The fifth and final robustness check involved flagging. Toward the end of March 2020, Twitter began censoring and flagging COVID-related fake news. To control for this, we removed all observations starting two weeks before March end. Online Supplemental Appendix 2 Table 2.4 shows that our results for the restricted sample remain statistically significant and consistent for all main variables. Flagging may have been effective but did not change our conclusions.

## Discussion and Implications

### Discussion

Unlike earlier studies that considered all corrections homogeneous, our study identified distinct subpopulations with varying effects on the diffusion of fake news. We also investigated different dimensions of information updates on a much larger scale, interactions with media files, and their effect on the probability of effective corrections. This section elaborates on our key findings.

The positive effect of *quality* updates on the probability of effective corrections confirms that updates of corrections with increasing uses of self-references and including provisions such as links and sources in support of their assertions are more likely to influence believability and reduce fake news diffusion as opposed to those that do not. A possible explanation could be that self-referential language in updated corrections promotes honesty, trust, and believability [20], which increases its probability of being shared. An established correlation exists in the literature between authenticity, reflecting self-referential language, and credibility [46, 47]. A preponderance of evidence supporting the correction claim fosters even more confidence in its truth as receivers can verify the veracity of the correction statement with other sources. Including additional sources, especially those perceived as credible, will motivate receivers to reevaluate the article by extending cognitive effort as posited by ELM [93]. However, corrective statements that use fewer self-references while being factual might not be as effective, as they may be met with skepticism and thus unlikely to be believed. As the authenticity of the correction updates increases relative to the original, their effectiveness also increases.

Our findings suggest that *quantity* updates increase the effectiveness of corrections. A possible explanation is that an increased word count per sentence could potentially equate with additional information. As the number of words per sentence increases in subsequent correction messages, the amount of shared information also increases. In addition, because rumors are more potent when information is lacking during times of uncertainty [2], more information boosts credibility [5] for correction messages. It increases the resharing of the corrections while reducing the believability of fake news and its diffusion. Our results are consistent with belief updates [28, 53] and the Bayes Theorem [7], where users are more likely to update their prior beliefs and reconcile incoherent probabilities as more information becomes available. By contrast, purely corrective updates, devoid of supplemental information, could necessitate additional information-seeking

behavior on the recipient's part, rendering them less effective than those with increasing amounts of additional information. This is possibly because people could be further exposed to inaccuracies during their information gathering, increasing the likelihood of disseminating them. Note that a paucity of information often fuels rumors [2]. Thus, people must receive additional information as it becomes available for corrections to be effective.

Our results also show that *relation* updates increase the chances of effective correction. Correction updates that deflect from the original and provide an alternative causal explanation of the event can fill the information gap left behind by the corrected misinformation instead of one that does not. Previous research indicates that the pervasive impact of misinformation can be mitigated by offering a counter-narrative explaining its inaccuracies [52] without repeating the false claim. Conversely, correction updates that repeat the original corrections may not provide alternative accounts but repeat instances of the fake news, thus potentially amplifying the misinformation. The original corrections from fact-checking organizations often simply refute the fake news and then reference the fake news they are correcting. As a result, subsequent correction updates that repeat these corrections without providing an alternative account are more likely to increase the likelihood of familiarity and reinforce it. These repetitions during corrections may, paradoxically, enhance the fake news story's memorability, thereby bolstering the misconception it aims to rectify [76]. The familiarity backfire effect manifests when individuals exhibit increased conviction in misinformation, often due to repeated exposure to the misinformation in corrections [78]. Thus, corrections that appear to repeat the original corrections verbatim may also repeat the fake news, which may negatively affect the correction process [89]. In addition, attempts to correct entirely novel misinformation by repeating falsehoods in a correction risk disseminating it to a wider audience that is previously unfamiliar with the false claims [22]. These findings provide fertile ground for future research.

The estimates further indicate that *manner* updates reduce the likelihood of an effective correction. The level of detail (not vagueness) provided relative to the original correction posts follows the law of diminishing marginal utility. The finding is not unsurprising, as pathological liars rely on excessive or elaborate details in verbal and nonverbal communications [41, 49] and often update lies with even more details over time to make the stories credible [100]. This is more likely to negatively affect the correction message's credibility. While the negative coefficient of *manner* updates in the probability equation directly contradicts the empirical testing of IMT, it is consistent with the literature on pathological liars. This could reflect information overload or be explained by an aversion to complex explanations, which may generate an "overkill backfire effect" [52] and a lack of believability associated with communications from pathological liars. This is also consistent with an inverted U-shaped relationship, which may lead to a blowback effect when the level of detail is too large.

Consistent with the literature, our estimate aligns with research on persuasive messages and the use of warnings, which has shown that messages tend to be less persuasive when they are too detailed or elaborate [14, 25, 29]. Easily processed information enjoy an advantage, irrespective of the chosen criteria for truth assessment, because it is perceived as more familiar, common, logically consistent, and congruent with personal convictions. Alternatively, cognitively demanding information is more prone to elicit critical scrutiny. In the event of perceived incongruities and heightened scrutiny, an increased search for relevant information and a greater commitment to determining factual accuracy are

observed [72]. Our measures of *manner* and *quantity* are likely to be connected because *quantity* is captured using word count. To obtain *ceteris paribus* effects, the probability equation includes both factors to help assess the effect of one while controlling for the other.

We argue that typically, providing complex or technical jargon negatively affects engagement and reduces trust in the correction. The empirical support for H3 suggests that while additional information can enhance the efficacy of corrections, excessive detail may hinder it.

Finally, our results show that while simply resharing media files, which include images or videos, decreases the likelihood of effective correction messages, hashtags increase it. During persuasion, theorizing from the ELM [93], media files often serve as cues and appeal to peripheral information processing routes. Shin et al. [74] observed that propagators of falsehoods often piggyback on important issues by including a trending hashtag that helps with assimilation, suggesting a deliberate attempt by disinformation agents to leverage this strategy to amplify messages. With corrections, hashtags may have the opposite effect, as they can bring readers' attention to the fake news unless accompanied by a deflative correction. Additionally, including media files in corrections updates that use self-referents and provide external sources to support their assertions further increases the probability of their effectiveness. However, they have the opposite effect when the corrections include adequate information. A possible explanation could be that, the additional information is enough for believability resharing, but including media files to that correction type may cause information overload.

### **Theoretical Contributions**

This study makes several theoretical contributions to the literature. First, we extend fake news literature by overcoming the complexities of tracing real-world information. We achieve this using a large-scale exogenous shock captured by waves of multiple fake news topics during the onset of the COVID-19 pandemic. Although IS scholars have examined intervention mechanisms in controlled laboratory settings [21, 22], our large-scale dataset and identification strategy provide a prime opportunity to capture interactions and dynamics. Although some related studies on fake news have also used large-scale datasets [45, 64], their main research questions differed.

Second, most studies have treated the diffusion and interactions of fake news and corrections as static communication processes. Our contributions include granular dynamic changes and information updates across four dimensions. The results show that allowing for dynamics is crucial for understanding the effectiveness of the corrections.

Third, research has treated correction messages as homogeneous. Our results show that this is not only likely to suffer from omitted variable bias, but it also misses a subpopulation of correction types that effectively reduce fake news. Our pooled model, which considers homogeneous corrections, is ineffective at combating misinformation. The preferred specification of the two types identifies effective and ineffective corrections, labeled based on their effects on fake news. The results show that separating corrections between subpopulations is critical to understanding how to reduce the spread of fake news and provides a holistic view of how both types of intervention affect its spread. Furthermore, although counts are a good measure of information diffusion [44], we found evidence of information potency beyond counts.

Finally, to our knowledge, this is the first study to extend the use of IMT to analyze correction messages. We also investigate the inclusion of media files to understand their moderating effects on fake news. While studies have leveraged IMT to understand deceptive discourse [58, 90], using correction updates or including media files is novel. Our results on the effects of *quality*, *quantity*, *relation*, and *manner*, in addition to media and hashtags, show promising results in reducing falsehoods. While studies on deceptive messages exploit the assumptions of conversational interactants as they relate to all four maxims [58], we tested the theory to understand whether corrections and IMT were resistant to including media files and explored the interactions and nonlinearities.

### **Practical Implications**

Our findings offer several implications for social media platform administrators, policy-makers, fact-checking organizations, and government agencies. Social media administrators and fact-checking organizations can design more effective corrections by encouraging relevant correction updates but reducing the amount of detail and self-references. Platform interventions that rely on corrections are often ineffective [17, 45] and require more effort to understand how different correction designs work. Our study responds to this call by empirically examining the effects of two types of corrections: an ineffective type, less common but more potent and amplifies fake news, and an effective type, which is common but less potent and reduces fake news. Our intervention further identified the update characteristics that differentiated them and their effectiveness. The results suggest that social media administrators may want to refrain from combining excessive information with media files, which may cause information overload. However, including media files when using self-referents to correct fake news improves its efficacy but may not be helpful in convincing users not to share fake news, especially when corrections are deflective or detailed.

Our findings suggest a possible consequence of an increased echo chamber and herding behavior [4], where people often follow crowds for a period regardless of information exposure. Our empirical results suggest that herd mentality can propel *quality*, *quantity*, and *manner* updates to reduce fake news, but the effectiveness of *relation* updates is adversely affected by herding. When designing correction messages during disasters, social media administrators, policymakers, government agencies, and fact-checking organizations might rely on *quality*, *quantity*, and *manner* of updates for information to be propelled more widely by irrational behavior. This is especially important during emergencies when rapid information dissemination is vital. However, authorities should note that herd mentality can negatively affect corrections based on *relation* updates. Thus, deflection may not be an efficient way to spread messages quickly.

Another practical implication is that social media administrators and brand managers can leverage this study to understand better the different dimensions of updates and how they affect engagement. Emotions often influence reactivity, and emotionally charged messages are more likely to be shared than neutral messages, causing reactivity and changes in content [36, 43, 75]. This reactivity results from users' need for personalization, modifying messages to synchronize them with their current affective state. Thus, a deeper understanding of the different dimensions of updates provides an even more granular understanding of users' different emotional states and can be used as a proxy for setting

thresholds on appropriate toxicity levels in community platforms, encouraging a healthier social media ecosystem. Message updates may also be used as a proxy for user interest and early adoption. Adopting our identification strategy can help advertisers, social media content creators, politicians, and marketing and sales executives improve their marketing mix, thus positively impacting sales and retention. For example, getting people to react to a new product post and modifying the content may help developers determine how different groups react differently during the initial launch. If people update their posts without self-referents, they may be distancing themselves from the product (e.g., during political campaigns), pointing to a larger unobserved issue.

Additionally, mixture models are generalized techniques for consolidating complex distributional features into a unified probabilistic model with broad domain applicability. The spread of falsehoods during disasters diverts state, local, and federal resources from helping recover to refuting falsehoods. Our empirical approach can help organizations and platform administrators proactively remove spammers, fake news, bots, trolls, and artificial intelligence (AI) from their platforms, fostering a community of helpful comments. With the current surge in bots, spammers, and trolls on social media and e-commerce platforms, accurately detecting them is increasingly important for firms and their customers. The nature of our approach and its robustness to ever-evolving tactics to beat the system make it a useful tool for platform administrators. Social media platforms may struggle to obtain sufficiently accurate labeled data for model training. However, the prohibitive cost and resource demands of manual labeling and ongoing model parameter updates pose significant challenges. Our approach can be used to model complex groups in multivariate distributions, enabling platforms to detect groups of spammers, trolls, bots, and even AI tools in academic settings, thus making them healthier.

Finally, while providing users with information is important, being too detailed might be counterproductive. During disasters, falsehoods can have serious consequences, particularly in evacuation zones. People may decide not to evacuate despite warnings from the state and local emergency management, thus risking their and the first responders' lives. Social media administrators and policymakers should provide enough information to non-expert citizens but should not be overly detailed or technical to avoid information overload.

## Conclusion and Future Research Directions

Fake news undermines trust, exacerbates social and political divisions, and leads to the manipulation of key aspects of our health, economy, and democracy. This study contributes to our understanding of the ways to prevent the spread of fake news. Using a unique dataset, we separate the subpopulations of corrections that have differentiated effects on fake news diffusion. We identified corrections that effectively reduced fake news that were more prevalent but less potent. We also identified corrections that we labeled ineffective because they amplify fake news. These corrections are less common but more potent. We found *quality*, *quantity*, and *relation* information updates increased the probability of a correction message being effective. By contrast, increasing *manner* updates decreases this probability. Hashtags increased the probability of effective corrections, but media files decreased them.

Our study is complex and, thus, not devoid of limitations. First, a deeper analysis of the emotional elements of fake news and corrections is beyond the scope of this study. Although we identified the effects of the different dimensions of correction updates and

their effects on fake news, it would be beneficial to understand what motivates social media users to update their information. We provide evidence of herd mentality related to correction updates. We believe a more robust identification of human judgment and behavior factors that drive users to update their information is required. In addition, future research should investigate the role of different media content in correction updates and their effectiveness. IMT has often been used to investigate information manipulation in deceptive discourse. Although we investigated correction updates, investigating the manipulation of fake news was beyond the scope of our study. The effectiveness of these factors depends on the believability and cognitive emotions. Future studies should investigate how fake news and corrections are updated.

Factors such as polarization, cognitive dissonance, and bias can influence behavior. Users may update their beliefs based on cognitive bias, dissonance, contagion, or polarization effects. Future research should investigate these antecedents using an experimental lens. Another promising area of investigation is the role of IMT in contagion. While previous work has shown that governments and independent fact-checkers are often wrong in assessing the veracity of information, none of our topics were found to have been reversed. These reversals can be used as external shocks to understand their effects on reducing the propagation of fake news.

Towards the end of our data collection period, Twitter began censoring and flagging COVID-related fake news. While our results were robust to these flagging and mitigation efforts, it is unclear whether they had a subtler impact on fake news diffusion. Therefore, additional examination is required to assess the efficacy of censoring. Finally, although we tried to identify heterogeneous fake news topics, selecting specific topics during crisis events represents a boundary beyond which our findings must be applied carefully. The politically and economically motivated nature of most fake news [31, 70] limits the generalizability of our findings.

## Notes

1. An example is the FEMA Coronavirus Rumor Control Page (Coronavirus Rumor Control | FEMA.gov).
2. They comprised nine business students from a large public university in the southern United States and nine education students from a private university in the northeast. Of these, 12 PhD students were involved in topic and keyword selection. By contrast, six master's students were involved in data cleaning, keyword querying, preprocessing, and labelling. We also received help from eight undergraduate students during the data collection. The original data consisted of over 200 million observations, took over three years to complete, and comprised six crisis events and included 6 months of COVID-19 data from January 2020 to June 2020; 4 weeks of Hurricane Florence data from September 7 to October 4, 2018; 5 weeks of Hurricane Harvey data from August 17 to September 27, 2017; the Ebola pandemic data of 2014; the Nepal Earthquake data of 2015; and the Fort McMurray Wildfire data of 2016.
3.  $(0.191/1000 \times 48) \times 100$  percent = 0.917 percent.

## Disclosure statement

No potential conflict of interest was reported by the author(s).

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